The Invariance Principle — problem 1.7

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There are a white, b black, and c red chips on a table. In one step, you may choose two chips of different colors and replace each one by a chip of the third color. Find conditions for all chips to become of the same color. Suppose you have initially 13 white 15 black and 17 red chips. Can all chips become of the same color ? What states can be reached from these numbers ?